Paper 12 - CA Prediction Surface

CQE-paper-12
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Construction Basis

Convert digital-physics and cellular-automaton candidates into prediction surfaces tied to local rules.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-12.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-12\paper_kernel_manifest.json
- body: production\papers\CQE-paper-12\PAPER-BODY.md
- source: production\papers\CQE-paper-12\SOURCE.md
- formal: production\papers\CQE-paper-12\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-12\02-CQE-tool\TOOL.md
- workbook: production\papers\CQE-paper-12\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-12.md

Paper 12 - CA Prediction Surface

Abstract

Paper 12 defines the prediction surface for deterministic binary cellular automata. A prediction surface is not a single magic predictor. It is a typed layer stack in which each layer reports its input, output, cost class, defect, and receipt.

The closed result in this paper is finite and local. For Rule 30, the local truth table agrees with:

```
Rule30(L,C,R) = L xor (C or R)
```

and with the local readout law:

```
T_EMISSION(L,C,R) = not L if C=1 else L xor R
```

The same local rule decomposes as:

```
Rule30(L,C,R) = Rule90(L,R) xor (C and not R)
```

The paper also verifies that exactly 64 of the 256 elementary cellular automata have silent boundary values `f(000)=0` and `f(111)=0`. The paper does not prove a cold-start full `O(log N)` Rule 30 extractor and does not prove the spectral next-state layer.

Claims

Claim 12.1. The Rule 30 local truth table matches `L xor (C or R)` on all eight LCR states.

Claim 12.2. The `T_EMISSION` formula matches Rule 30 on all eight LCR states.

Claim 12.3. Rule 30 decomposes as `Rule90 xor (C and not R)` on all eight LCR states.

Claim 12.4. Exactly 64 of the 256 elementary cellular automata are silent-boundary rules.

Claim 12.5. A prediction surface must preserve layer, cost, defect, and receipt metadata; empirical or open layers cannot be counted as closed.

Definitions

An **elementary cellular automaton** is a radius-1 binary rule:

```
f : {0,1}^3 -> {0,1}
```

For rule number `r`, the local emission is:

```
emit_r(L,C,R) = (r >> (4L + 2C + R)) & 1
```

A **prediction surface** is:

```
surface(system, N) -> (bit, layer, cost, defect, receipt)
```

For Rule 30:

```
Rule30(L,C,R) = L xor (C or R)
Rule90(L,R)   = L xor R
correction    = C and not R
```

A **silent-boundary rule** is an ECA satisfying:

```
emit_r(0,0,0) = 0
emit_r(1,1,1) = 0
```

Theorem 12.1 - CA Prediction Surface Finite Local Layers

The Rule 30 local readout, `T_EMISSION`, and Rule90-correction decomposition are exact on the eight binary LCR states. The 256-rule ECA space contains exactly 64 silent-boundary rules. These facts form the closed finite layer of the CA prediction surface.

Proof

Enumerate all states:

```
(L,C,R) in {0,1}^3
```

For each state, compute `emit_30(L,C,R)` from the ECA rule number and compute `L xor (C or R)`. The verifier checks equality on all eight rows.

For `T_EMISSION`, if `C=1` then:

```
L xor (C or R) = L xor 1 = not L
```

If `C=0` then:

```
L xor (C or R) = L xor R
```

Therefore `T_EMISSION` matches Rule 30 on every local state.

For the correction decomposition:

```
C or R = C xor R xor (C and R)
```

so:

```
Rule30 = L xor C xor R xor (C and R)
       = (L xor R) xor (C xor (C and R))
       = Rule90 xor (C and not R)
```

The silent-boundary count is finite. Two of the eight truth-table entries are fixed to zero. The remaining six entries are free, giving:

```
2^6 = 64
```

silent-boundary rules.

Receipt

The primary executable receipt is:

```
production/formal-papers/CQE-paper-12/verify_ca_prediction_surface.py
production/formal-papers/CQE-paper-12/ca_prediction_surface_receipt.json
```

The receipt status is `pass`. It verifies:

```
rule30_truth_table_matches_formula        = true
t_emission_matches_rule30                 = true
rule30_rule90_correction_identity_holds   = true
local_state_count_is_8                    = true
silent_boundary_rule_count_is_64          = true
cold_start_rule30_extractor_left_as_obligation = true
spectral_prediction_left_as_empirical     = true
```

Falsifiers

The verifier rejects:

```
the spectral layer is a proved cold-start Rule 30 predictor
```

```
a local T_EMISSION receipt proves between-depth dynamics without the local state
a layer may omit its cost and defect receipt
```

Role in the Suite

Paper 11 admits candidate theories through a receipt gate. Paper 12 turns an admitted cellular automaton candidate into a prediction surface:

```
admission receipt
-> local CA rule
-> layer stack
-> closed local readout
-> open extraction or empirical layers where required
```

Paper 13 may use the CA correction field as a quark-face transport input, but must preserve which layer produced the bit and which obligations remain open.

Open Audit Boundaries

- A full cold-start `O(log N)` Rule 30 extractor remains open.
- The spectral next-state layer is empirical until an accuracy theorem is supplied.
- Case-by-case closure of every silent-boundary rule requires additional receipts beyond the count.
- Any use of a prediction surface must preserve layer, cost, defect, and receipt metadata.

Conclusion

Paper 12 proves the finite local layer of the cellular-automaton prediction surface. It gives Rule 30 an exact local readout and correction decomposition, counts the silent-boundary ECA subset, and keeps cold-start and spectral prediction claims visible as open layers rather than hidden conclusions.